



Coral islands

The tropical western Pacific has thousands of coral-fringed islands. Some of these islands are the rocky summits of extinct oceanic volcanoes, but others are formed of coral sand. Built by living organisms, the coral reefs support all kinds of wildlife, and are the richest of all marine habitats.

◀ HIDDEN TREASURE

The clear tropical sea around this sandy island conceals a wealth of underwater life living among the luxuriant reef corals.

Submerged seamounts

Only a fraction of the volcanoes that have erupted from the ocean floor are visible as islands. Most of them form submerged mountains known as seamounts. Some of these were once volcanic islands that became extinct and sank below sea level. Others are still active and growing. Many are in long chains, including the Emperor Seamount chain that stretches across the ocean for more than 3,730 miles (6,000 km).

▼ VITAL SUPPLIES

The ocean currents swirling up and over hidden seamounts carry vital food to the surface, and support marine life such as these manta rays.



Shrinking ocean

In some places, the Pacific floor is expanding from spreading rifts that create mid-ocean ridges, such as the East Pacific Rise near South America. Some parts of the ocean floor move faster than others, making the rock crack along the sliding faults of fracture zones. But the earthquake zones around the Pacific are destroying ocean floor faster than it is created, so the ocean is shrinking.

